

ZAMYN-UUD, Mongolia

WMO: 443580

Lat: 43.703N

Lon: 111.901E

Elev: 964

StdP: 90.27

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.6	-26.4	-32.1	0.2	-26.0	-30.0	0.3	-24.7	12.0	-7.5	10.2	-11.6	1.3	90	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	34.5	19.2	32.7	18.5	31.0	17.9	22.1	30.8	20.8	29.3	19.7	27.5	3.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	15.9	26.1	18.1	14.6	24.8	17.0	13.7	24.2	70.0	31.1	65.0	29.5	61.1	27.6	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	9.9	8.9	-30.8	37.2	2.4	1.9	-32.5	38.5	-33.9	39.6	-35.3	40.6	-37.0	42.0
			-30.9	22.9	2.4	2.3	-32.7	24.6	-34.1	26.0	-35.5	27.3	-37.3	29.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-17.1	-11.6	-2.0	7.7	14.9	21.1	24.4	22.2	15.7	6.1	-5.3	-14.4		
	DBStd	15.09	5.04	5.89	5.77	5.38	4.73	3.67	3.02	3.24	4.33	5.02	6.06	5.09		
	HDD10.0	3297	841	604	374	105	11	0	0	0	7	139	459	756		
	HDD18.3	5239	1099	838	632	320	126	15	1	6	100	379	709	1014		
	CDD10.0	1553	0	0	1	36	164	334	445	379	176	18	0	0		
	CDD18.3	454	0	0	0	1	21	99	187	126	19	0	0	0		
	CDH23.3	5012	0	0	0	32	344	1111	2048	1242	229	4	0	0		
	CDH26.7	2007	0	0	0	6	112	433	931	469	57	0	0	0		
	WSAvg	3.1	2.6	2.9	3.5	4.0	4.2	3.2	2.8	2.7	3.0	3.1	3.0	2.8		
	PrecAvg	116	1	1	2	5	10	19	30	27	11	5	3	2		
Precipitation	PrecMax	254	15	6	11	23	55	68	168	80	42	36	19	8		
	PrecMin	50	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	47	3	1	3	6	11	13	28	19	9	7	4	2		
	PrecStd	47	3	1	3	6	11	13	28	19	9	7	4	2		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.2	7.4	18.1	27.1	32.7	35.5	37.7	35.4	30.9	23.6	12.8	2.5		
		MCWB	-3.6	2.0	6.6	12.9	15.6	18.3	20.5	20.6	16.2	13.1	5.5	-1.9		
	2%	DB	-3.8	3.8	13.7	23.3	29.8	33.2	35.2	33.0	28.3	20.2	9.3	-1.8		
		MCWB	-6.5	-0.9	5.1	11.0	15.0	17.9	19.8	19.4	15.7	11.3	3.6	-4.7		
	5%	DB	-6.3	1.1	10.8	20.4	27.1	31.1	33.3	31.1	25.9	17.5	6.4	-4.4		
		MCWB	-8.4	-2.8	3.3	9.5	13.6	17.3	19.2	18.4	14.6	9.3	1.5	-6.7		
	10%	DB	-8.8	-1.6	7.8	17.7	24.7	29.1	31.5	29.2	23.8	14.8	3.7	-6.5		
		MCWB	-10.4	-4.6	1.4	8.1	12.6	16.9	18.6	17.7	13.9	7.5	-0.2	-8.3		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.6	2.3	7.8	14.3	18.5	21.6	24.3	23.4	19.3	14.2	6.5	-1.5		
		MCDB	-0.5	7.4	16.1	24.5	30.0	31.0	33.8	32.1	27.4	22.1	12.1	2.4		
	2%	WB	-6.4	-0.8	5.6	11.9	16.2	20.3	22.5	21.8	17.4	11.9	3.7	-4.5		
		MCDB	-3.9	3.4	13.3	21.3	26.2	29.8	31.7	30.1	25.0	19.0	9.1	-1.8		
	5%	WB	-8.3	-2.6	3.5	10.1	14.5	19.2	21.2	20.4	16.1	9.7	1.8	-6.5		
		MCDB	-6.4	1.0	10.4	19.1	24.5	28.0	29.7	28.3	23.4	16.4	6.3	-4.4		
	10%	WB	-10.4	-4.6	1.6	8.4	13.3	18.1	20.1	19.2	14.7	8.0	0.1	-8.2		
		MCDB	-8.8	-1.7	7.6	17.1	23.0	26.3	27.9	26.3	22.1	14.2	3.8	-6.4		
Mean Daily Temperature Range	5% DB	MDBR	11.8	13.8	14.0	13.7	13.7	12.2	11.5	11.6	12.6	12.7	11.9	10.8		
		MCDBR	14.1	17.0	17.5	17.4	17.6	15.1	14.2	14.0	15.2	16.1	14.4	12.7		
		MCWBR	11.9	13.0	11.5	10.5	9.5	7.0	5.5	6.0	7.5	9.7	10.1	10.5		
	5% WB	MCDBR	14.0	16.5	17.1	16.1	15.8	13.0	11.9	12.2	13.0	15.4	14.0	12.7		
		MCWBR	12.0	12.7	11.7	10.5	9.5	7.3	6.0	6.5	7.8	10.0	10.1	10.6		
Clear-Sky Solar Irradiance	taub		0.270	0.301	0.342	0.393	0.438	0.466	0.436	0.421	0.382	0.335	0.289	0.271		
	taud		2.283	2.200	2.126	2.031	1.959	1.982	2.130	2.138	2.175	2.241	2.280	2.255		
	Ebn at Noon		859	884	888	866	835	812	833	831	835	832	823	814		
	Edh at Noon		83	109	135	162	179	176	150	144	128	104	83	78		
All-Sky Solar Radiation	RadAvg		2.17	3.31	4.96	6.24	6.88	6.66	6.59	6.09	5.08	3.78	2.56	1.85		
	RadStd		0.26	0.25	0.31	0.30	0.35	0.39	0.27	0.23	0.20	0.19	0.19	0.24		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	-0.99	N/A	-0.53	-0.73	N/A	N/A	N/A	N/A

Nomenclature: See separate page